Abelian Subgroup Covers Under a Bound on Pairwise Noncommuting Sets

Abstract

For a group G, let $\omega_{\mathrm{nc}}(G)$ denote the largest cardinality of a set of pairwise noncommuting elements, and let $\mathrm{ac}(G)$ denote the least number of abelian subgroups whose union is G. We study

$$h(n) = \sup\{\mathrm{ac}(G) : \omega_{\mathrm{nc}}(G) \le n\}.$$

A completely elementary argument gives

$$2^{\lfloor (n-1)/2 \rfloor} + 1 \le h(n) \le U_n \qquad (n \ge 3),$$

where $U_1 = U_2 = 1$ and $U_n = nU_{n-2}$ for $n \ge 3$. Thus

$$U_{2r} = 2^{r-1}r!, \qquad U_{2r+1} = (2r+1)!!,$$

and consequently

$$\exp\left(\frac{\log 2}{2}n - O(1)\right) \le h(n) \le \exp\left(\frac{1}{2}n\log n - \frac{1}{2}n + O(\log n)\right).$$

The lower bound is realized by explicit extraspecial 2-groups: for every $r \ge 1$ we construct a group E_r satisfying

$$\omega_{\mathrm{nc}}(E_r) = 2r+1, \qquad \mathrm{ac}(E_r) = 2^r + 1.$$

We also prove $h(1) = h(2) = 1$, $h(3) = 3$, and $h(4) = 4$, and show that $h(n) \ge n$ for every even $n \ge 4$. Finally, invoking the quantitative centre theorem of Neumann–Pyber, one obtains the sharp order of growth

$$h(n) = 2^{\Theta(n)}.$$

All arguments other than the two explicitly cited centre theorems of B. H. Neumann and Pyber are proved in full.

1 Definitions and statement of the result

Throughout, groups need not be finite. Covering and chromatic numbers are understood as cardinal numbers until finiteness is proved.

Definition 1.1. Let G be a group. Its *noncommuting number* is the cardinal

$$\omega_{\mathrm{nc}}(G) = \sup\big\{|X| : X \subseteq G \text{ and } xy \ne yx \text{ for all distinct } x, y \in X\big\}.$$

Its *abelian covering number* is the least cardinal

$$\mathrm{ac}(G) = \min\left\{|\mathcal{A}| : \mathcal{A} \text{ is a family of abelian subgroups of } G \text{ and } G = \bigcup_{A \in \mathcal{A}} A\right\}.$$

Such a family always exists, for example the family of cyclic subgroups $\{\langle g \rangle : g \in G\}$. For $n \ge 1$, set

$$h(n) = \sup\{\mathrm{ac}(G) : G \text{ a group and } \omega_{\mathrm{nc}}(G) \le n\}.$$

The elementary upper bound proved below shows that this supremum is a finite integer; it is therefore exactly the minimal integer asked for in the problem.

The hypothesis in the problem is exactly $\omega_{\mathrm{nc}}(G) \le n$: a subset with no commuting pair is precisely a pairwise noncommuting subset.

Define a sequence $(U_n)_{n\ge1}$ by

$$U_1 = U_2 = 1, \qquad U_n = nU_{n-2} \quad (n \ge 3). \tag{1}$$

Theorem 1.2 (Main theorem)**.** *The following assertions hold.*

(i) *For every $n \ge 3$,*

$$2^{\lfloor (n-1)/2 \rfloor} + 1 \le h(n) \le U_n. \tag{2}$$

(ii) *For $r \ge 1$,*

$$U_{2r} = 2^{r-1} r!, \qquad U_{2r+1} = (2r+1)!! = \frac{(2r+1)!}{2^r r!}. \tag{3}$$

Consequently,

$$\frac{\log 2}{2} n - O(1) \le \log h(n) \le \frac{1}{2} n \log n - \frac{1}{2} n + O(\log n). \tag{4}$$

(iii) *For every even $n \ge 4$, one also has $h(n) \ge n$.*

(iv) *The first four values are*

$$h(1) = h(2) = 1, \qquad h(3) = 3, \qquad h(4) = 4. \tag{5}$$

(v) *There is an absolute constant $C > 1$ such that*

$$h(n) \le C^n \qquad (n \ge 1). \tag{6}$$

In particular,

$$h(n) = 2^{\Theta(n)}.$$

Parts (i)–(iv) will be proved entirely from first principles. Part (v) will be deduced from the two external centre theorems stated explicitly in §9; the reduction from their published forms to the precise form needed here is proved in full.

2 The noncommuting graph

Definition 2.1. The *noncommuting graph* $\Gamma(G)$ of a group G is the simple graph with vertex set G in which two distinct vertices are adjacent exactly when they do not commute.

Thus $\omega_{\mathrm{nc}}(G) = \omega(\Gamma(G))$, where ω denotes clique number.

Proposition 2.2. *For every group G,*

$$\mathrm{ac}(G) = \chi(\Gamma(G)),$$

where either side is allowed to be infinite.

Proof. Suppose first that

$$G = A_1 \cup \cdots \cup A_k$$

with each A_i an abelian subgroup. For every $g \in G$, choose one index $i(g)$ for which $g \in A_{i(g)}$, and color g with color $i(g)$. If two elements receive the same color, then they lie in one abelian subgroup and therefore commute. Hence adjacent vertices of $\Gamma(G)$ receive different colors, so

$$\chi(\Gamma(G)) \le k.$$

Taking the minimum over all abelian subgroup covers gives

$$\chi(\Gamma(G)) \le \mathrm{ac}(G).$$

Conversely, suppose that $\Gamma(G)$ has a proper coloring with color classes S_i. Every two elements of a fixed S_i commute. We claim that the subgroup $\langle S_i \rangle$ is abelian. Indeed, every element of $\langle S_i \rangle$ is a finite product of integral powers of elements of S_i; since all elements of S_i commute pairwise, any two such products may be reordered term by term and hence commute. Thus $\langle S_i \rangle$ is abelian. Since the color classes cover G, the subgroups $\langle S_i \rangle$ cover G. Therefore

$$\mathrm{ac}(G) \le \chi(\Gamma(G)).$$

Combining the two inequalities proves the proposition. □

The proposition is conceptually useful: the problem asks for a chromatic bound in terms of clique number, but only for the special class of noncommuting graphs of groups.

3 Centralizer lemmas

For $x \in G$, write

$$C_G(x) = \{g \in G : gx = xg\}, \qquad Z(G) = \bigcap_{x \in G} C_G(x).$$

Lemma 3.1 (A nonabelian group contains a noncommuting triple)**.** *If $x, y \in G$ do not commute, then the three elements*

$$x, \qquad y, \qquad xy$$

are pairwise noncommuting. Consequently, if $\omega_{\mathrm{nc}}(G) \le 2$, then G is abelian.

Proof. By assumption, x and y do not commute. If x commuted with xy, then

$$x(xy) = (xy)x.$$

Cancelling one copy of x on the left gives $xy = yx$, a contradiction. Similarly, if y commuted with xy, then

$$y(xy) = (xy)y,$$

and right cancellation of y would again give $yx = xy$. Hence the three displayed elements are pairwise noncommuting.

If G were nonabelian, there would be noncommuting $x, y \in G$, and the first part would give a pairwise noncommuting set of cardinality 3. Thus $\omega_{\mathrm{nc}}(G) \le 2$ forces G to be abelian. □

The next lemma is the key point in the elementary upper bound. It improves the obvious loss of one unit to a loss of two units.

Lemma 3.2 (Centralizer drop by two)**.** *Let G be a group with $\omega_{\mathrm{nc}}(G) < \infty$, and let $x \in G \setminus Z(G)$. Then*

$$\omega_{\mathrm{nc}}(C_G(x)) \le \omega_{\mathrm{nc}}(G) - 2.$$

Proof. Set $m = \omega_{\mathrm{nc}}(G)$. Let $T \subseteq C_G(x)$ be an arbitrary pairwise noncommuting set. Since x is not central, choose $y \in G$ such that $xy \neq yx$.

For each $t \in T$, at least one of t and xt fails to commute with y. Indeed, if both t and xt commuted with y, then both would lie in the subgroup $C_G(y)$, and hence

$$x = (xt)t^{-1} \in C_G(y),$$

contrary to the choice of y. Choose $\varepsilon_t \in \{0,1\}$ such that

$$u_t = x^{\varepsilon_t} t$$

does not commute with y.

We first verify that the elements u_t, $t \in T$, are pairwise noncommuting. If $s, t \in T$ are distinct, then $s, t \in C_G(x)$, and therefore

$$u_s u_t = x^{\varepsilon_s + \varepsilon_t} st, \qquad u_t u_s = x^{\varepsilon_s + \varepsilon_t} ts.$$

Since $st \neq ts$, it follows that $u_s u_t \neq u_t u_s$.

Each u_t commutes with x, because both x and t commute with x. Consequently,

$$\begin{aligned} u_t(xy) = (xy)u_t &\iff x u_t y = x y u_t \\ &\iff u_t y = y u_t. \end{aligned}$$

The final condition is false by the choice of u_t. Thus every u_t fails to commute with both y and xy.

Finally, y and xy do not commute. This follows either from Lemma 3.1, or directly: if $y(xy) = (xy)y$, then cancellation gives $yx = xy$.

It follows that

$$\{u_t : t \in T\} \cup \{y, xy\}$$

is a pairwise noncommuting subset of G. Therefore

$$|T| + 2 \leq m.$$

Since T was arbitrary, $\omega_{\mathrm{nc}}(C_G(x)) \leq m - 2$. □

Lemma 3.3 (A maximal noncommuting set covers by centralizers)**.** *Let $X = \{x_1, \ldots, x_m\}$ be a maximal, with respect to inclusion, pairwise noncommuting subset of a group G. Then*

$$G = \bigcup_{i=1}^{m} C_G(x_i).$$

In particular, the conclusion holds when X has maximum possible cardinality.

Proof. Let $g \in G$. If g failed to commute with every x_i, then

$$X \cup \{g\}$$

would be a strictly larger pairwise noncommuting set, contradicting the maximality of X. Hence $g \in C_G(x_i)$ for at least one i. □

The following strengthening is not required for the elementary recurrence, but it identifies the centre as the common centralizer of a maximum noncommuting set.

Lemma 3.4 (Common centralizer of a maximum set)**.** *Let $X = \{x_1, \ldots, x_m\}$ be a pairwise noncommuting subset of maximum possible cardinality in G. Then*

$$\bigcap_{i=1}^{m} C_G(x_i) = Z(G).$$

Proof. The inclusion

$$Z(G) \subseteq \bigcap_{i=1}^{m} C_G(x_i)$$

is immediate. Suppose, for a contradiction, that there exists

$$z \in \bigcap_{i=1}^{m} C_G(x_i) \setminus Z(G).$$

Choose $y \in G$ such that $zy \neq yz$.

For each i, at least one of x_i and zx_i fails to commute with y. Otherwise both would lie in the subgroup $C_G(y)$, and then

$$z = (zx_i)x_i^{-1} \in C_G(y),$$

a contradiction. Choose $\varepsilon_i \in \{0, 1\}$ such that

$$u_i = z^{\varepsilon_i} x_i$$

does not commute with y.

Because z commutes with every x_i, for $i \neq j$ one has

$$u_i u_j = z^{\varepsilon_i+\varepsilon_j} x_i x_j, \qquad u_j u_i = z^{\varepsilon_i+\varepsilon_j} x_j x_i.$$

The elements x_i and x_j do not commute, so neither do u_i and u_j. Thus

$$\{y, u_1, \ldots, u_m\}$$

is a pairwise noncommuting set of size $m+1$, contradicting the maximality of $|X| = m$. Therefore the common centralizer is exactly $Z(G)$. □

4 The elementary upper bound

We first record the elementary properties of the recurrence (1).

Lemma 4.1. *The sequence (U_n) is nondecreasing and satisfies, for every $r \geq 1$,*

$$U_{2r} = 2^{r-1} r!, \qquad U_{2r+1} = (2r+1)!! = \frac{(2r+1)!}{2^r r!}.$$

Moreover,

$$\log U_n = \frac{1}{2} n \log n - \frac{1}{2} n + O(\log n).$$

Proof. Iterating the recurrence separately on the two parity classes gives

$$U_{2r} = (2r)(2r-2)\cdots 4\, U_2 = \prod_{j=2}^{r} 2j = 2^{r-1} r!,$$

$$U_{2r+1} = (2r+1)(2r-1)\cdots 3\, U_1 = (2r+1)!!.$$

The standard identity

$$(2r+1)!! = \frac{(2r+1)!}{2^r r!}$$

follows by separating the even and odd factors in $(2r+1)!$.

It remains to check monotonicity across adjacent parity classes. Put

$$A_r = \frac{U_{2r+1}}{U_{2r}}, \qquad B_r = \frac{U_{2r+2}}{U_{2r+1}}.$$

One has $A_1 = 3 > 1$ and $B_1 = 4/3 > 1$. Using the recurrence,

$$\frac{A_{r+1}}{A_r} = \frac{2r+3}{2r+2} > 1, \qquad \frac{B_{r+1}}{B_r} = \frac{2r+4}{2r+3} > 1.$$

Thus $A_r > 1$ and $B_r > 1$ for every r, proving that $U_{n+1} \geq U_n$.

Finally, the logarithmic form of Stirling's formula,

$$\log(r!) = r \log r - r + O(\log(r+1)),$$

applied to the two explicit formulas above yields

$$\log U_n = \frac{1}{2} n \log n - \frac{1}{2} n + O(\log n).$$

□

Theorem 4.2 (Elementary covering theorem)**.** *Let G be a group with $\omega_{\mathrm{nc}}(G) = m < \infty$. Then*

$$\mathrm{ac}(G) \leq U_m.$$

Consequently, $h(n) \leq U_n$ for every $n \geq 1$.

Proof. We argue by strong induction on m.

If $m \leq 2$, Lemma 3.1 shows that G is abelian. Hence $\mathrm{ac}(G) = 1 = U_m$.

Now assume that $m \geq 3$ and that the assertion is known for every smaller noncommuting number. Choose a maximum pairwise noncommuting set

$$X = \{x_1, \ldots, x_m\}.$$

By Lemma 3.3,

$$G = \bigcup_{i=1}^{m} C_G(x_i).$$

Every x_i is noncentral, because it fails to commute with every x_j for $j \neq i$. Hence Lemma 3.2 gives

$$\omega_{\mathrm{nc}}(C_G(x_i)) \leq m - 2.$$

By the induction hypothesis and the monotonicity of U_n,

$$\mathrm{ac}(C_G(x_i)) \leq U_{\omega_{\mathrm{nc}}(C_G(x_i))} \leq U_{m-2}.$$

Thus each of the m centralizers is a union of at most U_{m-2} abelian subgroups. Taking all these subgroup covers together gives

$$\mathrm{ac}(G) \leq m U_{m-2} = U_m.$$

This completes the induction.

If $\omega_{\mathrm{nc}}(G) \leq n$, then monotonicity gives

$$\mathrm{ac}(G) \leq U_{\omega_{\mathrm{nc}}(G)} \leq U_n.$$

Taking the supremum over such groups proves $h(n) \leq U_n$. □

Corollary 4.3. *As $n \to \infty$,*

$$h(n) \leq \exp\left(\frac{1}{2} n \log n - \frac{1}{2} n + O(\log n)\right).$$

Proof. Combine Theorem 4.2 with Lemma 4.1. □

5 Symplectic linear algebra over the field of two elements

The lower-bound construction will be phrased in symplectic language. We include all required linear algebra.

Definition 5.1. Let W be a vector space over $\mathbb{F}_2$. A bilinear form

$$B : W \times W \longrightarrow \mathbb{F}_2$$

is *alternating* if $B(w, w) = 0$ for every $w \in W$, and is *nondegenerate* if the only vector orthogonal to every vector of W is 0. A subspace $L \leq W$ is *totally isotropic* if $B(u, v) = 0$ for all $u, v \in L$.

For a subspace $L \leq W$, write

$$L^{\perp} = \{w \in W : B(w, \ell) = 0 \text{ for every } \ell \in L\}.$$

Lemma 5.2 (Symplectic basis)**.** *Let (W, B) be a finite-dimensional nondegenerate alternating space over $\mathbb{F}_2$. Then $\dim W$ is even, say $\dim W = 2r$, and there exists a basis*

$$e_1, \ldots, e_r, f_1, \ldots, f_r$$

such that

$$B(e_i, e_j) = B(f_i, f_j) = 0, \qquad B(e_i, f_j) = \delta_{ij}.$$

Consequently, any two nondegenerate alternating spaces over $\mathbb{F}_2$ of the same finite dimension are isometric.

Proof. We use induction on $\dim W$. The zero-dimensional case is immediate. Suppose $W \neq 0$, and choose $e_1 \neq 0$. Nondegeneracy gives $f_1 \in W$ with $B(e_1, f_1) = 1$. Let

$$H = \operatorname{span}\{e_1, f_1\}.$$

The restriction of B to H is nondegenerate. Indeed, if $ae_1 + bf_1$ is orthogonal to both e_1 and f_1, then

$$0 = B(ae_1 + bf_1, e_1) = bB(f_1, e_1) = b,$$

and

$$0 = B(ae_1 + bf_1, f_1) = aB(e_1, f_1) = a.$$

Here $B(f_1, e_1) = B(e_1, f_1) = 1$, because in characteristic 2 an alternating bilinear form is symmetric: from

$$0 = B(u + v, u + v) = B(u, v) + B(v, u)$$

one obtains $B(u, v) = B(v, u)$.

The linear map

$$W \longrightarrow H^*, \qquad w \longmapsto \big(h \mapsto B(w, h)\big),$$

is surjective because its restriction to H is an isomorphism. Therefore $\dim H^{\perp} = \dim W - 2$. Since $H \cap H^{\perp} = \{0\}$, one has

$$W = H \oplus H^{\perp}.$$

The restriction of B to $H^{\perp}$ is nondegenerate: if $w \in H^{\perp}$ is orthogonal to all of $H^{\perp}$, then it is orthogonal to both summands in the displayed direct sum, hence to all of W, so $w = 0$.

The induction hypothesis applied to $H^{\perp}$ supplies the remaining pairs e_i, f_i. This proves both the existence of a symplectic basis and the evenness of $\dim W$.

Finally, a linear map taking one symplectic basis to another preserves all pairings of basis vectors, and hence, by bilinearity, preserves the form on the whole space. Thus spaces of the same dimension are isometric. $\square$

Lemma 5.3 (Dimension of an isotropic subspace). *Let (W, B) be a $2r$-dimensional nondegenerate alternating space over $\mathbb{F}_2$. If $L \leq W$ is totally isotropic, then*

$$\dim L \leq r.$$

Proof. The linear map

$$\Phi : W \longrightarrow L^*, \qquad \Phi(w)(\ell) = B(w, \ell),$$

is surjective. To see this, choose a basis of L and extend it to a basis of W; equivalently, use nondegeneracy to identify W with W^* and then restrict linear functionals from W to L. Its kernel is exactly $L^\perp$. Rank–nullity therefore gives

$$\dim L^\perp = 2r - \dim L.$$

Since L is totally isotropic, $L \subseteq L^\perp$. Hence

$$\dim L \leq \dim L^\perp = 2r - \dim L,$$

which is equivalent to $\dim L \leq r$. □

Lemma 5.4 (Rank of the off-diagonal all-ones matrix). *Let M_k be the $k \times k$ matrix over $\mathbb{F}_2$ whose diagonal entries are 0 and whose off-diagonal entries are 1. Then*

$$\operatorname{rank} M_k = \begin{cases} k, & k \text{ even}, \\ k-1, & k \text{ odd}. \end{cases}$$

Proof. Let $\mathbf{1} = (1, \ldots, 1)^T$, and for $u = (u_1, \ldots, u_k)^T$ put

$$s = \sum_{i=1}^{k} u_i \in \mathbb{F}_2.$$

Since $M_k = I_k + J_k$, where J_k is the all-ones matrix,

$$M_k u = u + s\mathbf{1}.$$

Thus $M_k u = 0$ if and only if $u = s\mathbf{1}$. Consistency with the definition of s requires

$$s = \sum_{i=1}^{k} s = ks$$

in $\mathbb{F}_2$. If k is even, this forces $s = 0$, so the kernel is trivial and the rank is k. If k is odd, both $s = 0$ and $s = 1$ are possible, and the kernel is precisely $\operatorname{span}\{\mathbf{1}\}$; hence the rank is $k - 1$. □

6 An explicit family giving the exponential lower bound

Fix $r \geq 1$. Put

$$V = \mathbb{F}_2^r \oplus \mathbb{F}_2^r.$$

For $a, b, c, d \in \mathbb{F}_2^r$, define

$$B((a, b), (c, d)) = a \cdot d + b \cdot c, \tag{7}$$

where $\cdot$ denotes the usual dot product over $\mathbb{F}_2$. The form B is alternating, because

$$B((a, b), (a, b)) = a \cdot b + b \cdot a = 0,$$

and nondegenerate, because orthogonality to all $(0, d)$ forces $a = 0$, while orthogonality to all $(c, 0)$ forces $b = 0$.

6.1 The group and its commutation law

Let

$$E_r = V \times \mathbb{F}_2.$$

Write elements as triples (a, b, ε) with $a, b \in \mathbb{F}_2^r$ and $\varepsilon \in \mathbb{F}_2$, and define multiplication by

$$(a, b, \varepsilon)(c, d, \eta) = (a + c,\ \ b + d,\ \ \varepsilon + \eta + a \cdot d). \tag{8}$$

Proposition 6.1. *The operation* (8) *makes E_r a group of order 2^{2r+1}. The map*

$$\pi : E_r \longrightarrow V, \qquad \pi(a, b, \varepsilon) = (a, b),$$

is a surjective homomorphism, and

$$Z(E_r) = \ker \pi = \{(0, 0, \varepsilon) : \varepsilon \in \mathbb{F}_2\}.$$

Moreover, for $g, h \in E_r$,

$$gh = hg \quad \Longleftrightarrow \quad B(\pi(g), \pi(h)) = 0. \tag{9}$$

Proof. Let $u = (a, b)$, $v = (c, d)$, and $w = (e, f)$ in V, and set

$$\kappa(u, v) = a \cdot d.$$

Bilinearity gives the cocycle identity

$$\begin{aligned} \kappa(u, v) + \kappa(u + v, w) &= a \cdot d + (a + c) \cdot f \\ &= a \cdot d + a \cdot f + c \cdot f \\ &= c \cdot f + a \cdot (d + f) \\ &= \kappa(v, w) + \kappa(u, v + w). \end{aligned}$$

This identity is exactly the equality of the third coordinates in

$$((u, \varepsilon)(v, \eta))(w, \theta) \quad \text{and} \quad (u, \varepsilon)((v, \eta)(w, \theta)),$$

so the multiplication is associative.

The element $(0, 0, 0)$ is an identity. A direct calculation shows that

$$(a, b, \varepsilon)^{-1} = (a, b, \varepsilon + a \cdot b),$$

because all vector coordinates are in characteristic 2. Thus E_r is a group. Its underlying set has $2^r \cdot 2^r \cdot 2 = 2^{2r+1}$ elements.

The formula (8) immediately shows that π is a surjective homomorphism and that $\ker \pi$ is central. For

$$g = (a, b, \varepsilon), \qquad h = (c, d, \eta),$$

the first two coordinates of gh and hg agree, while their third coordinates are respectively

$$\varepsilon + \eta + a \cdot d \quad \text{and} \quad \varepsilon + \eta + c \cdot b.$$

Hence

$$gh = hg \iff a \cdot d + c \cdot b = 0 \iff B((a, b), (c, d)) = 0,$$

which proves (9).

If $g \notin \ker \pi$, then $\pi(g) \neq 0$. By nondegeneracy of B, there exists $v \in V$ such that $B(\pi(g), v) = 1$. Any lift $h \in E_r$ of v fails to commute with g by (9). Therefore no element outside $\ker \pi$ is central, and $Z(E_r) = \ker \pi$. $\square$

6.2 The exact noncommuting number

Theorem 6.2. *For every $r \geq 1$,*

$$\omega_{\mathrm{nc}}(E_r) = 2r + 1.$$

Proof. We first prove the upper bound. Let

$$g_1, \ldots, g_k \in E_r$$

be pairwise noncommuting, and put $v_i = \pi(g_i) \in V$. By (9),

$$B(v_i, v_j) = 1 \qquad (i \neq j).$$

In particular, the v_i are distinct: if $v_i = v_j$, then $B(v_i, v_j) = B(v_i, v_i) = 0$.

Choose a basis of V, let S be the matrix of B in that basis, and let A be the $2r \times k$ matrix whose ith column is the coordinate vector of v_i. The Gram matrix of $v_1, \ldots, v_k$ is

$$A^T S A = M_k,$$

where M_k is the matrix from Lemma 5.4. Therefore

$$\operatorname{rank} M_k = \operatorname{rank}(A^T S A) \leq \operatorname{rank} S = 2r.$$

If k is even, Lemma 5.4 gives $k \leq 2r$. If k is odd, it gives $k - 1 \leq 2r$, and hence $k \leq 2r + 1$. Thus in all cases

$$k \leq 2r + 1.$$

For the matching lower bound, consider

$$W = \left\{ u = (u_1, \ldots, u_{2r+1}) \in \mathbb{F}_2^{2r+1} : \sum_{i=1}^{2r+1} u_i = 0 \right\}$$

with the ordinary dot product. The space W has dimension $2r$. The restricted dot product is alternating, since for $u \in W$,

$$u \cdot u = \sum_i u_i^2 = \sum_i u_i = 0.$$

It is nondegenerate. Indeed, in the ambient space the orthogonal complement of W is the one-dimensional subspace generated by

$$\mathbf{1} = (1, \ldots, 1),$$

because $W = \mathbf{1}^{\perp}$. Since $2r + 1$ is odd,

$$\sum_{i=1}^{2r+1} 1 = 1 \quad \text{in } \mathbb{F}_2,$$

so $\mathbf{1} \notin W$. Hence

$$W \cap W^{\perp} = \{0\},$$

which is exactly nondegeneracy of the restricted form.

For $1 \leq i \leq 2r + 1$, let

$$w_i = \mathbf{1} + e_i,$$

where e_i is the ith standard basis vector. The vector w_i has exactly $2r$ nonzero coordinates, so $w_i \in W$. If $i \neq j$, then w_i and w_j are simultaneously nonzero in exactly

$$(2r + 1) - 2 = 2r - 1$$

coordinates. Therefore

$$w_i \cdot w_j = 1 \quad \text{in } \mathbb{F}_2.$$

By Lemma 5.2, the nondegenerate alternating space W is isometric to (V, B). Transporting the vectors w_i through an isometry gives vectors

$$v_1, \ldots, v_{2r+1} \in V$$

with $B(v_i, v_j) = 1$ for $i \neq j$. Choosing arbitrary lifts $g_i \in E_r$ of the v_i, criterion (9) shows that the g_i are pairwise noncommuting. Hence

$$\omega_{\mathrm{nc}}(E_r) \geq 2r + 1.$$

Together with the upper bound, this proves equality. □

6.3 A symplectic spread

To determine $\mathrm{ac}(E_r)$ exactly, we need a partition of the nonzero vectors of V into maximal totally isotropic subspaces.

Lemma 6.3 (Existence of $\mathbb{F}_{2^r}$)**.** *For every $r \geq 1$, there exists a field F with exactly 2^r elements. As a vector space over $\mathbb{F}_2$, it has dimension r.*

Proof. Let $q = 2^r$, and let K be a splitting field over $\mathbb{F}_2$ of

$$P(X) = X^q - X.$$

Its derivative is

$$P'(X) = qX^{q-1} - 1 = 1$$

in characteristic 2, so P has no repeated roots. Since P splits in K and has degree q, its root set

$$F = \{x \in K : x^q = x\}$$

has exactly q elements.

The set F is a field. It contains 0 and 1. If $x, y \in F$, then repeated use of the Frobenius identity $(a + b)^2 = a^2 + b^2$ gives

$$(x + y)^q = x^q + y^q = x + y,$$

so $x + y \in F$. Also

$$(xy)^q = x^q y^q = xy,$$

so $xy \in F$. If $x \in F$ is nonzero, then $x^{q-1} = 1$, and therefore

$$(x^{-1})^q = (x^q)^{-1} = x^{-1};$$

hence $x^{-1} \in F$. Thus F is a field with $q = 2^r$ elements.

If $d = [F : \mathbb{F}_2]$, then F has 2^d elements as a d-dimensional vector space over $\mathbb{F}_2$. Therefore $2^d = 2^r$, so $d = r$. □

Proposition 6.4 (Symplectic spread)**.** *Every $2r$-dimensional nondegenerate alternating space over $\mathbb{F}_2$ possesses $2^r + 1$ totally isotropic r-dimensional subspaces whose pairwise intersections are $\{0\}$ and whose union is the whole space.*

Proof. Let $F = \mathbb{F}_{2^r}$, whose existence is supplied by Lemma 6.3. Choose a nonzero $\mathbb{F}_2$-linear functional

$$\lambda : F \longrightarrow \mathbb{F}_2.$$

On the $2r$-dimensional $\mathbb{F}_2$-vector space F^2, define

$$\beta((x,y),(x',y')) = \lambda(xy' + x'y).$$

This form is bilinear and alternating. It is nondegenerate: if $(x,y) \neq (0,0)$ and $x \neq 0$, choose $z \in F$ with $\lambda(z) = 1$ and take $(x',y') = (0, x^{-1}z)$; then

$$\beta((x,y),(x',y')) = 1.$$

If $x = 0$, then $y \neq 0$, and taking $(x',y') = (y^{-1}z, 0)$ gives the same conclusion.

For each $t \in F$, define

$$L_t = \{(u, tu) : u \in F\},$$

and define

$$L_\infty = \{(0, u) : u \in F\}.$$

Each L_t and L_∞ has dimension r over $\mathbb{F}_2$. Moreover, for $u, v \in F$,

$$\beta((u,tu),(v,tv)) = \lambda(utv + vtu) = \lambda(0) = 0,$$

so L_t is totally isotropic; L_∞ is visibly totally isotropic as well.

If $s \neq t$, then a vector in $L_s \cap L_t$ satisfies

$$(u, su) = (u, tu),$$

so $(s-t)u = 0$. Since F is a field and $s - t \neq 0$, this forces $u = 0$. Thus $L_s \cap L_t = \{0\}$. Similarly, $L_t \cap L_\infty = \{0\}$.

Finally, every nonzero $(x,y) \in F^2$ belongs to one of these subspaces. If $x = 0$, it lies in L_∞. If $x \neq 0$, then

$$(x,y) = (x, tx) \quad \text{with} \quad t = yx^{-1},$$

so it lies in L_t. The preceding intersection calculation also shows uniqueness.

Thus the $2^r + 1$ subspaces

$$\{L_t : t \in F\} \cup \{L_\infty\}$$

partition the nonzero vectors of F^2. By Lemma 5.2, any given $2r$-dimensional nondegenerate alternating space is isometric to (F^2, β); transporting the displayed family through an isometry proves the proposition. □

6.4 The exact abelian covering number

Theorem 6.5. *For every $r \geq 1$,*

$$\mathrm{ac}(E_r) = 2^r + 1.$$

Proof. We first establish the lower bound. Let $A \leq E_r$ be abelian. Since π is a homomorphism, $\pi(A)$ is a linear subspace of V. If $u, v \in \pi(A)$, choose $g, h \in A$ with $\pi(g) = u$ and $\pi(h) = v$. Since A is abelian, $gh = hg$, and criterion (9) gives

$$B(u,v) = 0.$$

Thus $\pi(A)$ is totally isotropic. By Lemma 5.3,

$$\dim \pi(A) \leq r,$$

and consequently

$$|\pi(A) \setminus \{0\}| \leq 2^r - 1.$$

Suppose now that

$$E_r = A_1 \cup \cdots \cup A_k$$

with each A_i abelian. Then the subspaces $\pi(A_i)$ cover V: for each $v \in V$, the element $(v, 0) \in E_r$ lies in some A_i, so $v \in \pi(A_i)$. Counting nonzero vectors gives

$$2^{2r} - 1 = |V \setminus \{0\}| \leq \sum_{i=1}^{k} |\pi(A_i) \setminus \{0\}| \leq k(2^r - 1).$$

Therefore

$$k \geq \frac{2^{2r} - 1}{2^r - 1} = 2^r + 1.$$

Hence $\mathrm{ac}(E_r) \geq 2^r + 1$.

For the reverse inequality, take a symplectic spread

$$\mathcal{L} = \{L_1, \ldots, L_{2^r+1}\}$$

of V supplied by Proposition 6.4. For every $L \in \mathcal{L}$, the full inverse image

$$A_L = \pi^{-1}(L)$$

is a subgroup of E_r. It is abelian, because for $g, h \in A_L$ one has $\pi(g), \pi(h) \in L$, and total isotropy gives

$$B(\pi(g), \pi(h)) = 0;$$

criterion (9) then yields $gh = hg$.

Since the members of $\mathcal{L}$ cover V, their inverse images cover E_r. Thus E_r is covered by $2^r + 1$ abelian subgroups, and

$$\mathrm{ac}(E_r) \leq 2^r + 1.$$

Together with the lower bound, this proves equality. □

Corollary 6.6 (Exponential lower bound)**.** *For every $n \geq 3$,*

$$h(n) \geq 2^{\lfloor (n-1)/2 \rfloor} + 1.$$

Consequently,

$$\log h(n) \geq \frac{\log 2}{2} n - O(1).$$

Proof. Set

$$r = \left\lfloor \frac{n-1}{2} \right\rfloor.$$

Then $r \geq 1$ and

$$\omega_{\mathrm{nc}}(E_r) = 2r + 1 \leq n$$

by Theorem 6.2. Therefore the definition of $h(n)$ and Theorem 6.5 give

$$h(n) \geq \mathrm{ac}(E_r) = 2^r + 1.$$

Since

$$\left\lfloor \frac{n-1}{2} \right\rfloor \geq \frac{n}{2} - 1,$$

the logarithmic estimate follows. □

Combining Corollaries 4.3 and 6.6 proves the completely elementary estimate (4).

7 A dihedral lower bound for even parameters

The symplectic construction is asymptotically stronger, but a dihedral family gives the useful exact lower bound $h(n) \geq n$ for every even $n \geq 4$.

Proposition 7.1. *Let $q \geq 3$ be odd, and let*

$$D_{2q} = \langle r, t : r^q = t^2 = 1,\ trt = r^{-1} \rangle$$

be the dihedral group of order $2q$. Then

$$\omega_{\mathrm{nc}}(D_{2q}) = q + 1, \qquad \mathrm{ac}(D_{2q}) = q + 1.$$

Proof. Every element of D_{2q} has a unique form r^i or $r^i t$, with $i \in \mathbb{Z}/q\mathbb{Z}$. The q elements $r^i t$ are the reflections. For $i, j \in \mathbb{Z}/q\mathbb{Z}$,

$$(r^i t)(r^j t) = r^{i-j}, \qquad (r^j t)(r^i t) = r^{j-i}.$$

Thus two reflections commute exactly when

$$r^{2(i-j)} = 1.$$

Since q is odd, multiplication by 2 is invertible modulo q, so this condition forces $i = j$. Hence distinct reflections are pairwise noncommuting.

Let r^k be a rotation and $r^i t$ a reflection. Then

$$r^k (r^i t) = r^{k+i} t, \qquad (r^i t) r^k = r^{i-k} t.$$

They commute exactly when $r^{2k} = 1$, which, again because q is odd, forces $k = 0$. Therefore every nonidentity rotation fails to commute with every reflection.

It follows that the q reflections together with one nonidentity rotation form a pairwise noncommuting set of size $q + 1$. Conversely, a pairwise noncommuting set can contain at most one rotation, because all rotations lie in the cyclic subgroup $\langle r \rangle$ and commute with one another; it can contain at most all q reflections. Hence

$$\omega_{\mathrm{nc}}(D_{2q}) = q + 1.$$

For the covering number, an abelian subgroup contains at most one reflection, since distinct reflections do not commute. Moreover, an abelian subgroup containing a reflection contains no nonidentity rotation, because no such rotation commutes with a reflection. Thus covering all q reflections requires at least q abelian subgroups, and none of those subgroups covers a nonidentity rotation. At least one further abelian subgroup is therefore needed, so

$$\mathrm{ac}(D_{2q}) \geq q + 1.$$

On the other hand,

$$D_{2q} = \langle r \rangle \cup \bigcup_{i \in \mathbb{Z}/q\mathbb{Z}} \langle r^i t \rangle,$$

a union of one cyclic rotation subgroup and q cyclic reflection subgroups. Hence $\mathrm{ac}(D_{2q}) \leq q + 1$, proving equality. □

Corollary 7.2. *If $n \geq 4$ is even, then $h(n) \geq n$.*

Proof. Put $q = n - 1$. Then $q \geq 3$ is odd, and Proposition 7.1 gives

$$\omega_{\mathrm{nc}}(D_{2q}) = q + 1 = n, \qquad \mathrm{ac}(D_{2q}) = q + 1 = n.$$

The definition of $h(n)$ therefore yields $h(n) \geq n$. □

8 The exact values for the first four parameters

Theorem 8.1. *One has*

$$h(1) = h(2) = 1, \qquad h(3) = 3, \qquad h(4) = 4.$$

Proof. If $\omega_{\mathrm{nc}}(G) \le 2$, Lemma 3.1 shows that G is abelian. Hence

$$h(1) = h(2) = 1.$$

For $n = 3$, Theorem 4.2 gives

$$h(3) \le U_3 = 3.$$

Taking $r = 1$ in Theorems 6.2 and 6.5 gives a group E_1 with

$$\omega_{\mathrm{nc}}(E_1) = 3, \qquad \mathrm{ac}(E_1) = 3.$$

Thus $h(3) \ge 3$, and therefore $h(3) = 3$.

For $n = 4$, Theorem 4.2 gives

$$h(4) \le U_4 = 4.$$

Taking $q = 3$ in Proposition 7.1 gives the dihedral group $D_6 \cong S_3$ with

$$\omega_{\mathrm{nc}}(D_6) = 4, \qquad \mathrm{ac}(D_6) = 4.$$

Thus $h(4) \ge 4$, and hence $h(4) = 4$. □

9 The sharp exponential order

The elementary recurrence proves finiteness and gives an explicit superexponential upper bound. To obtain the correct exponential order, we use two established group-theoretic results. They are the only external structural inputs in this manuscript.

Theorem 9.1 (B. H. Neumann). *If a group G satisfies $\omega_{\mathrm{nc}}(G) < \infty$, then*

$$[G : Z(G)] < \infty.$$

Theorem 9.2 (Pyber, finite form). *There exists an absolute constant $C > 1$ such that every finite group Q satisfies*

$$[Q : Z(Q)] \le C^{\omega_{\mathrm{nc}}(Q)}.$$

Theorem 9.1 is due to B. H. Neumann, while Theorem 9.2 is Pyber's quantitative strengthening in the finite case; see [1, 2]. We next give all details needed to pass from these statements to arbitrary groups.

Lemma 9.3 (Schreier's finite-generation lemma). *If H is a finitely generated group and $K \le H$ has finite index, then K is finitely generated.*

Proof. Let S be a finite symmetric generating set for H, and choose a finite right transversal T for K in H with $1 \in T$. For $h \in H$, let $\overline{h} \in T$ be the unique representative satisfying

$$Kh = K\overline{h}.$$

For $t \in T$ and $s \in S$, define the Schreier element

$$\sigma(t, s) = ts\,\overline{ts}^{-1} \in K.$$

There are only finitely many such elements.

Let $k \in K$, and write

$$k = s_1 s_2 \cdots s_m \qquad (s_j \in S).$$

Put

$$t_j = \overline{s_1 s_2 \cdots s_j} \quad (0 \le j \le m),$$

where $t_0 = 1$. Since $k \in K$, one also has $t_m = 1$. A telescoping multiplication gives

$$\begin{aligned} k &= (t_0 s_1 t_1^{-1})(t_1 s_2 t_2^{-1}) \cdots (t_{m-1} s_m t_m^{-1}) \\ &= \sigma(t_0, s_1)\sigma(t_1, s_2) \cdots \sigma(t_{m-1}, s_m). \end{aligned}$$

Thus the finite set of Schreier elements generates K. $\square$

Lemma 9.4 (Separating finitely many elements in a finitely generated abelian group)**.** *Let A be a finitely generated abelian group, and let $S \subseteq A \setminus \{1\}$ be finite. There exists a finite-index subgroup $N \le A$ such that*

$$N \cap S = \varnothing.$$

Proof. By the structure theorem for finitely generated abelian groups,

$$A \cong \mathbb{Z}^d \oplus T$$

for some finite abelian group T. Fix $s \in S$. If the T-component of s is nontrivial, projection to T is a homomorphism to a finite group that does not annihilate s. Otherwise, some coordinate of the $\mathbb{Z}^d$-component of s is a nonzero integer a. Choose a prime p not dividing a and compose the corresponding coordinate projection with reduction modulo p; again this is a homomorphism to a finite group that does not annihilate s.

Thus, for every $s \in S$, there is a homomorphism

$$\varphi_s : A \longrightarrow F_s$$

to a finite group such that $\varphi_s(s) \ne 1$. Set

$$N = \bigcap_{s \in S} \ker \varphi_s.$$

This is a finite-index subgroup of A, because it is the kernel of the map from A to the finite direct product $\prod_{s \in S} F_s$. By construction, no member of S belongs to N. $\square$

Lemma 9.5 (Reduction to a finite group)**.** *Let G be a group for which $[G : Z(G)] < \infty$. Then there exists a finite group Q such that*

$$[Q : Z(Q)] = [G : Z(G)] \qquad \textit{and} \qquad \omega_{\mathrm{nc}}(Q) \le \omega_{\mathrm{nc}}(G).$$

Proof. Choose finitely many elements $g_1, \dots, g_d \in G$ whose cosets generate the finite quotient $G/Z(G)$, and put

$$H = \langle g_1, \dots, g_d \rangle.$$

Then

$$G = HZ(G).$$

We claim that

$$Z(H) = H \cap Z(G). \tag{10}$$

The inclusion from right to left is immediate. Conversely, if $z \in Z(H)$, then z commutes with H and also with $Z(G)$. Since $G = HZ(G)$, it follows that z commutes with all of G, so $z \in Z(G)$. This proves (10). Consequently,

$$[H : Z(H)] = [G : Z(G)] < \infty. \tag{11}$$

The group H is finitely generated, and $Z(H)$ has finite index in H. Lemma 9.3 therefore shows that the abelian group $Z(H)$ is finitely generated.

Choose a complete set R of representatives for the nonidentity cosets of $Z(H)$ in H. For each $t \in R$, choose $y_t \in H$ such that

$$c_t = [t, y_t] \neq 1;$$

such a choice is possible because $t \notin Z(H)$. Let

$$S = \{c_t : t \in R \text{ and } c_t \in Z(H)\}.$$

This is a finite subset of $Z(H) \setminus \{1\}$. By Lemma 9.4, there is a finite-index subgroup $N \leq Z(H)$ such that $N \cap S = \varnothing$. Since N is central in H, it is normal in H.

Set

$$Q = H/N.$$

The group Q is finite, because both $[H : Z(H)]$ and $[Z(H) : N]$ are finite. We claim that

$$Z(Q) = Z(H)/N. \tag{12}$$

Certainly $Z(H)/N \leq Z(Q)$. For the reverse inclusion, suppose that xN is central in Q. Write

$$x = tz$$

with $z \in Z(H)$ and with either $t = 1$ or $t \in R$. If $t \in R$, then

$$[x, y_t] = [tz, y_t] = [t, y_t] = c_t,$$

because z is central. Since xN is central in Q, this commutator must belong to N. If $c_t \notin Z(H)$, that is impossible because $N \leq Z(H)$. If $c_t \in Z(H)$, it is impossible because $c_t \in S$ and $N \cap S = \varnothing$. Hence $t = 1$, so $x \in Z(H)$. This proves (12).

Equations (11) and (12) give

$$[Q : Z(Q)] = [H/N : Z(H)/N] = [H : Z(H)] = [G : Z(G)].$$

Finally, if elements of Q are pairwise noncommuting, then arbitrary lifts to H are pairwise noncommuting: commuting lifts would have commuting images. Therefore

$$\omega_{\text{nc}}(Q) \leq \omega_{\text{nc}}(H) \leq \omega_{\text{nc}}(G).$$

□

Corollary 9.6 (Quantitative centre theorem in the required form)**.** *There exists an absolute constant $C > 1$ such that every group G with $\omega_{\text{nc}}(G) < \infty$ satisfies*

$$[G : Z(G)] \leq C^{\omega_{\text{nc}}(G)}.$$

Proof. By Theorem 9.1, $[G : Z(G)]$ is finite. Apply Lemma 9.5 to obtain a finite group Q with

$$[Q : Z(Q)] = [G : Z(G)] \quad \text{and} \quad \omega_{\text{nc}}(Q) \leq \omega_{\text{nc}}(G).$$

Theorem 9.2 then gives

$$[G : Z(G)] = [Q : Z(Q)] \leq C^{\omega_{\text{nc}}(Q)} \leq C^{\omega_{\text{nc}}(G)}.$$

□

Lemma 9.7. *If $[G : Z(G)] < \infty$, then*

$$\text{ac}(G) \leq [G : Z(G)].$$

Proof. Choose one representative g from each coset of $Z(G)$ in G. For such a representative, define

$$A_g = \langle g, Z(G) \rangle.$$

The subgroup A_g is abelian: every element of it has the form $g^k z$ with $k \in \mathbb{Z}$ and $z \in Z(G)$, and for any $k, \ell \in \mathbb{Z}$ and $z, w \in Z(G)$,

$$(g^k z)(g^\ell w) = g^{k+\ell} zw = g^{k+\ell} wz = (g^\ell w)(g^k z).$$

Moreover, A_g contains the entire coset $gZ(G)$. Since the cosets of $Z(G)$ cover G, the corresponding subgroups A_g cover G. Their number is exactly $[G : Z(G)]$. $\square$

Theorem 9.8. *There exists an absolute constant $C > 1$ such that*

$$h(n) \leq C^n$$

for every $n \geq 1$. Combined with Corollary 6.6, this gives

$$h(n) = 2^{\Theta(n)}.$$

Proof. Let G satisfy $\omega_{\mathrm{nc}}(G) \leq n$. By Corollary 9.6,

$$[G : Z(G)] \leq C^{\omega_{\mathrm{nc}}(G)} \leq C^n.$$

Lemma 9.7 therefore gives

$$\mathrm{ac}(G) \leq C^n.$$

Taking the supremum over all such groups yields $h(n) \leq C^n$.

On the other hand, Corollary 6.6 gives, for $n \geq 3$,

$$h(n) \geq 2^{\lfloor (n-1)/2 \rfloor} + 1 \geq 2^{n/2-1}.$$

Thus there are positive absolute constants c_1, c_2 such that

$$e^{c_1 n} \leq h(n) \leq e^{c_2 n}$$

for all sufficiently large n, which is precisely $h(n) = 2^{\Theta(n)}$. $\square$

10 Conclusion

The problem has a finite answer for every n. The fully explicit elementary bounds are

$$2^{\lfloor (n-1)/2 \rfloor} + 1 \leq h(n) \leq \begin{cases} 2^{r-1} r!, & n = 2r, \\ (2r+1)!!, & n = 2r+1, \end{cases} \qquad (n \geq 3).$$

In logarithmic form,

$$\frac{\log 2}{2} n - O(1) \leq \log h(n) \leq \frac{1}{2} n \log n - \frac{1}{2} n + O(\log n).$$

The quantitative centre theorem improves the upper bound to C^n, and hence the correct order of growth is exponential:

$$\boxed{h(n) = 2^{\Theta(n)}}.$$

The explicit symplectic family E_r shows that no subexponential upper bound is possible.

References

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[2] L. Pyber, *The number of pairwise noncommuting elements and the index of the centre in a finite group*, J. London Math. Soc. (2) **35** (1987), 287–295.